Survival driven deconvolution (DeSurv) reveals prognostic and interpretable cancer subtypes

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Bulk tumor transcriptomes reflect a mixture of malignant cells and microenvironmental components, and the gene programs that explain the most expression variance need not be the programs most associated with patient outcomes. Standard nonnegative matrix factorization (NMF) discovers latent structure by minimizing reconstruction error, which may be dominated by variance-dominant but prognostically neutral signals such as tissue composition. Here we present DeSurv, a survival-supervised deconvolution framework that jointly integrates NMF with Cox proportional hazards modeling, directing gene programs toward outcome-relevant structure during discovery rather than only in post hoc evaluation. In simulations with known latent structure, DeSurv more reliably recovers the true factorization rank and the identity of prognostic gene programs than standard NMF, with the largest advantage when variance and prognosis diverge. In pancreatic ductal adenocarcinoma (PDAC), DeSurv concentrates survival signal into a parsimonious set of factors aligned with known tumor and microenvironmental programs while suppressing variance-dominant exocrine-compositional signals. These survival-aligned factors generalize across independent PDAC cohorts with consistent hazard ratios and clearer survival separation than their unsupervised counterparts. These results support survival supervision as a principled approach for discovering interpretable, outcome-aligned transcriptional programs in heterogeneous tumors.

nonnegative matrix factorization | survival analysis | semi-supervised learning | tumor deconvolution | pancreatic cancer

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Predicting which cancer patients will respond to therapy or progress rapidly requires identifying the transcriptional signatures that drive clinical outcomes (1). Bulk RNA-sequencing remains the primary source of the large, clinically annotated cohorts required for such survival modeling, but each expression profile is a composite of malignant cells, cancer-associated fibroblasts, immune infiltrates, and other microenvironmental components (2). As a result, these signatures may be diluted or obscured by variation in cellular composition across samples. Nonnegative matrix factorization (NMF) has been instrumental in resolving this mixture into additive, interpretable gene programs corresponding to recognizable cell types or transcriptional states (3–5). This approach has yielded important biological insights, including the identification of basal-like and classical tumor programs in pancreatic ductal adenocarcinoma (PDAC) (1, 4) and compartment-specific deconvolution in pancreatic cancer (5).

However, the standard approach discovers these programs through unsupervised factorization and only then evaluates their clinical relevance retrospectively (6, 7). Because unsupervised NMF minimizes reconstruction error, its factors tend to be dominated by the highest-variance patterns in the data, which may reflect tissue composition or other outcome-neutral sources rather than prognostic biology (8, 9). In PDAC, exocrine and tissue-composition signals reflect low tumor purity rather than cancer cell-intrinsic programs and carry little prognostic value (10, 11). Although less prevalent in bulk tissue, the tumor-intrinsic basal-like and activated stromal programs carry the strongest survival associations (4), yet risk being diluted or missed entirely when decomposition prioritizes explaining maximal variance in expression. This misalignment between variance and prognosis extends beyond PDAC: tumor purity alone accounts for a large fraction of expression variation across cancer types (12), yet the objective optimized during unsupervised discovery (reconstruction error) differs fundamentally from the criterion used during evaluation (survival association) (9). Identifying the prognostically relevant subset then requires extensive downstream filtering that is ad hoc, cohort-specific, and difficult to reproduce (13). Over a decade of PDAC subtyping efforts proposed between two and six subtypes (1, 4, 13, 14), ultimately converging on a robust basal/classical dichotomy only after extensive retrospective evaluation across independent cohorts (10).

Whether incorporating survival information during factorization improves generalizability, or risks overfitting to cohort-specific outcome distributions, remains an open empirical question. Sufficient dimension reduction theory establishes that response-guided subspace estimation targets the directions most relevant to the outcome, whereas variance-maximizing projections can miss outcome-relevant structure entirely (9, 15). When the outcome depends on cell populations that

Significance Statement

Tumor transcriptomes mix malignant and microenvironmental signals, making it difficult to identify programs that drive clinical outcomes. Existing deconvolution and matrix factorization methods discover latent programs but do not ensure prognostic relevance, while supervised predictors often sacrifice biological interpretability. We present DeSurv, a survival-supervised deconvolution framework that integrates nonnegative matrix factorization with Cox modeling to learn gene programs and their survival associations jointly. By embedding outcome information into discovery and using automatic model selection, DeSurv yields clinically relevant, reproducible programs across cohorts. This advances tumor deconvolution and provides a general tool for identifying actionable drivers of disease progression.

A.M.Y. and N.U.R. conceived and designed the study; A.M.Y. developed the DeSurv method, implemented the software, performed all analyses, and generated figures; A.Y. and D.L. contributed to methodological development and analytic tools; A.M.Y. and N.U.R. wrote the manuscript; all authors reviewed and approved the final manuscript.

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are compositionally minor or contribute low variance to bulk expression, unsupervised methods systematically miss these features, while outcome-supervised methods can recover them (8, 15, 16). Supervision during factorization would filter out variance-dominant but prognostically neutral signals during learning rather than in a separate post hoc step, but whether this theoretical advantage translates to improved generalization in the NMF deconvolution setting, where nonnegative constraints, censored survival outcomes, and cohort-specific variation introduce additional challenges, has not been empirically tested.

Here we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox proportional hazards modeling. The key architectural choice is where survival supervision enters the factorization. In DeSurv, factor scores are defined as $Z = W^\top X$, so the Cox partial likelihood is a function of the gene program matrix W and regression coefficients β, and the survival gradient acts directly on gene programs. Sample-level loadings H are updated solely through the reconstruction objective and receive no survival gradient, preserving their interpretation as mixture coefficients (3, 17) and the biological interpretability of the deconvolution. DeSurv operates as a semi-supervised method: a parameter α balances reconstruction fidelity $(1-\alpha)$ and survival prediction (α). Because α is selected via cross-validated concordance, values large enough to overfit to cohort-specific survival patterns are penalized by poor out-of-sample performance, preventing the survival term from overwhelming the factorization. Because W defines shared transcriptomic programs, new samples can be scored by projection ($Z_{\text{new}} = W^\top X_{\text{new}}$) without requiring their survival data, a property not shared by methods that route supervision through H (18, 19). Hyperparameters, including factorization rank k and α, are jointly selected via cross-validated concordance (Methods).

We evaluate DeSurv in two settings that test the predictions above. First, in simulations with known latent structure and survival associations, we show that DeSurv recovers the true factorization rank and the identity of prognostic programs more reliably than standard NMF, that its advantage scales with the degree of divergence between variance and prognosis, and that it vanishes under null conditions where no survival signal exists. Second, in PDAC, we demonstrate that survival supervision reorganizes the learned factor structure: DeSurv suppresses variance-dominant but prognostically neutral signals (e.g., exocrine content) and concentrates survival association into a smaller set of biologically interpretable factors aligned with known tumor and microenvironmental programs. These survival-aligned factors generalize across independent PDAC cohorts with consistent hazard ratios and clearer survival separation than their unsupervised counterparts.

Results

Model Overview. We developed DeSurv, a survival-supervised deconvolution framework that jointly optimizes NMF reconstruction and Cox proportional hazards likelihood (Fig. 1; Methods). The survival gradient acts on the gene program matrix W but not on the sample loadings H, directing gene programs toward outcome-relevant structure while preserving interpretability of sample loadings as mixture coefficients (3, 17). We evaluated DeSurv in two settings: controlled simulations and PDAC cohort analysis with external validation.

Survival supervision clarifies NMF rank selection. Before examining DeSurv's performance across the three evaluation settings described above, we address a prerequisite question using both PDAC expression data and controlled simulations: whether incorporating clinical outcomes clarifies the well-documented instability of unsupervised rank selection heuristics (6, 20). Using gene expression data from the PDAC training cohorts (TCGA (21) and CPTAC (22); Methods), we first evaluated commonly used unsupervised criteria across a range of candidate ranks.

Standard NMF diagnostics yielded inconsistent guidance (Fig. 2A-C). Reconstruction residuals decreased smoothly with increasing k and did not exhibit a clear elbow, a pattern consistent with both relatively small solutions ($k \approx$ 3-4) and substantially larger ranks ($k \approx$ 6-8). The cophenetic correlation coefficient began to decline at low ranks ($k \approx$ 3-4) but continued to fluctuate at higher values without a distinct transition point. In contrast, mean silhouette width was highest at very small ranks ($k \approx$ 2-3) and decreased monotonically thereafter, favoring low-dimensional solutions that conflicted with the other criteria. Together, these unsupervised heuristics pointed to incompatible values of k, illustrating the ambiguity of rank selection in standard NMF.

To address this ambiguity, we applied DeSurv, which incorporates survival outcomes directly into the factorization and evaluates models using cross-validated concordance index (C-index). The resulting C-index surface across the joint space of factorization rank (k) and supervision strength (α) identified a well-defined optimum, in contrast to the conflicting recommendations produced by unsupervised heuristics (Fig. 2D). Model selection followed the one-standard-error rule: we selected the smallest k whose predicted performance lay within one standard error of the maximum, yielding a parsimonious choice. Bayesian optimization selected $k = 3$ and $\alpha = 0.334$ (C-index 0.655; 1-SE rule; $n = 273$ patients, 139 events).

To further evaluate rank recovery under controlled conditions, we conducted simulation studies in which the true underlying rank was known ($k = 3$). DeSurv selected the true rank ($k = 3$) more frequently than standard NMF and avoided the systematic under-selection seen in the unsupervised baseline, though variability across replicates remained (Fig. 2E). In contrast, standard NMF followed by post hoc Cox modeling ($\alpha = 0$) showed a dominant tendency to under-select, concentrating near $k = 2$. These results support the prediction that incorporating outcome information into model selection improves recovery of the true factorization rank.

DeSurv recovers prognostic gene programs when variance and prognosis diverge. Sufficient dimension reduction theory predicts that outcome-guided subspace estimation recovers directions most relevant to the response (9, 15), but this prediction has not been evaluated in the NMF deconvolution setting. To test whether it holds under nonnegative constraints with survival outcomes, we designed simulations with known ground-truth latent structure. Simulated expression matrices were generated from a nonnegative factor model in which prognostic gene programs explained low variance relative to outcome-neutral background signals, with survival times generated from marker gene expression ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Methods). We compared DeSurv to stan-

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dard NMF ($\alpha = 0$) followed by post hoc Cox regression, with both methods tuned via cross-validated concordance index using Bayesian optimization.

In the primary scenario, where prognostic programs explained low variance relative to outcome-neutral programs, DeSurv consistently achieved higher C-index and substantially improved precision—the fraction of genes in a learned factor that belong to a true prognostic gene program—for recovering the true prognostic gene programs (Fig. 3A-B). The unsupervised baseline exhibited near-zero precision, indicating that variance-driven factorization failed to concentrate on the genes that actually drove survival. This result supports the prediction that outcome-guided learning recovers prognostic programs more reliably when variance and prognosis diverge. Across 100 replicates, DeSurv achieved median C-index 0.839 versus 0.724 for standard NMF ($\Delta = 0.115$; $p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$).

To verify that these gains reflect genuine signal recovery rather than overfitting, we repeated the analysis under two additional simulation scenarios (SI Appendix, Fig. S2). In a null scenario where survival times were generated independently of gene expression ($\beta = 0$), DeSurv defaulted to standard NMF: Bayesian optimization selected low supervision strength (α), and both methods yielded C-index values near 0.5. This indicates that the semi-supervised design does not impose spurious prognostic structure when none exists. In a mixed scenario where survival depended on both factor-specific marker genes and shared background genes, creating partial overlap between variance-dominant and prognostically relevant structure, DeSurv's advantage was present but attenuated relative to the primary scenario. Together, these three scenarios establish that DeSurv's benefit scales with the degree of divergence between variance and prognosis: largest when prognostic programs explain low variance (primary scenario), moderate when variance and prognosis partially overlap (mixed), and absent when no survival signal exists (null).

Survival supervision reorganizes the learned factor structure in PDAC. In PDAC, standard NMF distributes its factors across variance-dominant but prognostically inert programs; DeSurv reorganizes the learned factor structure to concentrate survival signal into a single classical-iCAF axis.

To directly test whether survival supervision reorganizes the learned factor structure, as predicted by the simulation results above, we examined the overlap between factor-specific gene rankings and established PDAC gene programs (Fig. 4A-B). Bayesian optimization selected $k = 3$ and $\alpha = 0.334$ for DeSurv in the PDAC training cohorts (TCGA and CPTAC). To enable a direct comparison of how each method organizes the same number of factors, we also fit standard NMF at the same rank ($k = 3$); results for standard NMF at independently selected ranks ($k = 5$ via elbow detection, $k = 7$ via cross-validated concordance at $\alpha = 0$) are presented in the SI Appendix. We note that standard NMF at $k = 5$ achieves comparable external validation performance to DeSurv at $k = 3$; the DeSurv advantage lies in parsimony — recovering equivalent prognostic generalization with fewer, more interpretable factors — rather than in raw concordance at each method's unrestricted optimum.

DeSurv produced a factorization that concentrated survival signal into a single factor (D1) integrating classical tumor programs with an iCAF-associated stromal signature (Fig. 4A). This mirrors observations that survival stratification in PDAC is strongest when tumor-intrinsic and CAF subtypes are considered jointly, with classical tumors in an iCAF microenvironment demonstrating the most favorable outcomes (11, 23). A second factor (D2) captured an activated immune-stromal microenvironment. A third factor (D3) was enriched for basal-like tumor programs but, as discussed below, contributed minimally to survival in conditional analysis, reflecting both the absorption of residual exocrine-compositional variation and the concentration of the prognostically relevant tumor-stroma contrast within D1. Notably, no DeSurv factor showed a significant positive correlation with exocrine-associated gene programs (DECODER Exocrine, Collisson Exocrine, Bailey ADEX; Fig. 4A), suggesting that survival supervision deprioritizes differentiation-related variation that is weakly associated with outcome.

At the same rank, standard NMF organized the three factors around dominant sources of transcriptional variance rather than prognostic biology (Fig. 4B). One factor was strongly and significantly correlated with exocrine-associated gene programs (DECODER Exocrine, Collisson Exocrine, Bailey ADEX; Fig. 4B) and negatively correlated with immune and stromal signatures, consistent with a bulk composition axis separating normal or differentiated tissue from non-epithelial tumor content. A second factor aligned with classical tumor identity, while a third aggregated immune, fibroblast, and extracellular matrix-related programs — including Puleo Desmoplastic, SCISSORS panCAF and myCAF, Maurer Immune and ECM, and DECODER Activated Stroma — into a single composite microenvironmental factor (Fig. 4B). These patterns indicate that, given the same number of factors, standard NMF allocates substantial model capacity to differentiation-driven signals and merges distinct microenvironmental states that DeSurv separates.

We quantified these differences by contrasting the conditional fraction of expression variance explained by each factor with its survival contribution (SI Appendix, Fig. S7). A striking pattern emerged: all three NMF factors contributed negligibly to survival despite spanning a range of conditional variance, while DeSurv concentrated nearly all survival signal into D1. Specifically, the DeSurv factor with the largest survival contribution explained 7% of variance with $\Delta\ell = 66.4$, while the highest-variance NMF factor explained 23.9% of expression variance but contributed only $\Delta\ell = 0.1$ to survival. Notably, the highest-variance DeSurv factor (D3, ~30% of conditional variance) also contributed minimally to survival. This apparent paradox — given the well-established prognostic role of basal-like biology in PDAC (4, 10) — reflects two features of the factorization: D3 absorbs residual exocrine-compositional variation alongside basal-like programs (see factor correspondence below), diluting its prognostic specificity; and the survival-relevant contrast between classical and basal-like biology is already captured by D1's iCAF-classical co-program, leaving D3's conditional contribution minimal. This result directly illustrates the misalignment between variance and prognosis anticipated by sufficient dimension reduction theory (9) and observed empirically in tumor purity analyses (12): the factors that explain the most transcriptomic variation need not be the factors most associated with patient outcomes. The observed pattern in PDAC is consistent with the intermediate regime between the primary and mixed simulation

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scenarios, where DeSurv's advantage was largest.

To further characterize how the learned factor structure differs between DeSurv and standard NMF, we examined the W-matrix gene loading correspondence between factors derived from the two methods (SI Appendix, Fig. S8). NMF's classical-associated factor (N1) showed the strongest correspondence with DeSurv's basal-like factor (D3), and NMF's composite microenvironmental factor (N3) mapped primarily to DeSurv's microenvironmental factor (D2). The exocrine-like NMF factor (N2) did not correspond strongly to any single DeSurv factor; its signal was weakly absorbed into D3, consistent with D3's high conditional variance and low survival contribution described above. Together, these results demonstrate that DeSurv selectively preserves biologically distinct tumor and microenvironmental structure while reorganizing variance-dominant but prognostically inert variation — including exocrine-compositional signal — into a high-variance, low-survival factor rather than allocating dedicated model capacity to it.

Survival-aligned programs generalize across independent PDAC cohorts. Having shown that DeSurv reorganizes the learned factor structure in the training data, we next tested whether this learned factor structure generalizes to independent cohorts. If outcome-guided subspaces capture biologically reproducible rather than noise-driven structure, they should transfer more readily across datasets (9, 15). We evaluated whether DeSurv factors maintain their prognostic associations in independent PDAC cohorts. We computed factor scores in each validation dataset by projecting the gene programs learned in the training cohort ($Z = W^\top X_{\text{new}}$), yielding a continuous score for each factor per sample.

For each method (both fit at $k = 3$, the DeSurv BO-selected rank; external validation C-index for NMF at independently selected ranks is summarized in SI Appendix, Table S4), we focused on the factor showing the largest increase in Cox model log partial likelihood in the training data. Across 5 independent PDAC cohorts ($n = 616$), the DeSurv-derived factor exhibited consistent survival effects, with hazard ratio estimates showing limited variability and predominantly protective associations (continuous risk score model: pooled HR 1.78; 95% CI 1.48–2.13; $P = 4.4 \times 10^{-10}$; SI Appendix, Fig. S9). In contrast, the NMF factor identified by the same criterion showed greater heterogeneity across datasets and weaker survival associations, consistent with the expectation that variance-driven programs capture cohort-specific variation that does not transfer.

When validation samples were pooled and stratified using the cross-validated optimal cutpoint applied to the DeSurv linear predictor, clear separation of survival trajectories was observed (dichotomized high/low groups, Fig. 4C). The same procedure applied to NMF yielded weaker survival stratification (Fig. 4D). These results support the prediction that outcome-aligned programs capture biology that generalizes, whereas variance-driven programs may include cohort-specific signals that attenuate across datasets.

Factorization rank K=3 is robustly confirmed across supervision strength and gene-projection settings. The 1-SE model selection rule chose $k = 3$ over the global Bayesian optimization maximum ($k = 7$; CV C-index 0.655 versus 0.646 for $k = 3$; margin 0.009 C-index units). Without this rule, DeSurv would have selected $k = 7$ at $\alpha = 0.334$—the same rank independently favored by standard NMF under cross-validated concordance at $\alpha = 0$. The parsimony of the DeSurv solution relative to standard NMF therefore reflects a deliberate methodological choice encoded in the 1-SE rule, not an intrinsic property of survival supervision. Whether this preference is empirically warranted requires evaluating whether the additional complexity of $k = 7$ translates to improved generalization.

Bayesian optimization selected $k = 3$ and $\alpha = 0.334$ as the primary model configuration, but whether this reflects a stable solution or a fragile local optimum required further evaluation. To assess robustness, we conducted an exhaustive sensitivity analysis across $K \in \{2, 3, 5, 7, 9\}$ and $\alpha \in \{0, 0.25, 0.35, 0.55, 0.75, 0.85, 0.95\}$ (35 combinations), holding all remaining hyperparameters fixed at the production-optimal values ($\lambda = 0.349$, $\nu = 0.056$, ntop= 270, 100 initializations). For each fit, we identified the factor most correlated with the production K= 3 Factor 1 and assessed its survival association in 570 independent validation patients across 4 external cohorts (PACA-AU array and RNA-seq are treated as a single cohort here; the generalization analysis above counts them separately, yielding $n = 616$ across 5 datasets), using stratified Cox regression with and without adjustment for PurIST and DeCAF classifiers.

With the production-optimal gene-projection setting (ntop= 270), K= 3 validated adjustedly at 5 of 7 supervision-strength values, including $\alpha = 0.55$ where it achieved adjusted HR = 0.81 ($p = 0.003$) — nearly identical to the production fit ($p = 0.004$). K= 5 at $\alpha = 0.55$ provided independent corroboration at higher factor-program fidelity (Spearman H-score correlation = 0.903 vs. K= 3's 0.855; adjusted $p = 0.019$), confirming the iCAF biology is recoverable by a higher-rank model. In contrast, K= 7 — which had appeared to achieve adjusted significance ($p = 0.019$) in a parallel analysis using all 1970 genes in the projection — completely failed with ntop= 270 (adjusted $p = 0.676$), establishing that the earlier K= 7 advantage was an artifact of the all-genes projection rather than an intrinsic property of the 7-factor solution.

To isolate the contribution of gene focusing, we repeated the analysis with all genes retained (ntop= NULL) while keeping λ and ν at their optimal values. With all genes, K= 3 at $\alpha = 0.55$ reached only adjusted $p = 0.097$, failing to reproduce the production result, and only 5 of 35 combinations achieved adjusted $p < 0.05$ — compared with 18 of 35 with ntop= 270. Substituting $\lambda = 0.349$ for $\lambda = 0.300$ within the all-genes setting shifted the pattern of validated combinations but left the overall count unchanged (5/35), confirming that regularization tuning alone cannot reproduce the adjusted signal without gene focusing. These analyses establish ntop= 270 as the critical hyperparameter: focusing the projection onto the 270 most factor-characteristic genes concentrates the survival signal, suppresses noise from the remaining genes, and prevents overfitting at high supervision weights. Full K $\times$ α adjusted p-value tables are provided in the Supplementary (Tables S1–S3). Taken together, these results provide empirical justification for the 1-SE rule's parsimony preference: $k = 7$, which achieved the highest training CV C-index, completely failed adjusted external validation ($p = 0.676$), while $k = 3$ validated robustly ($p = 0.003$). The simpler model not only passed the 1-SE threshold—it generalized better.

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Discussion

We have shown that incorporating survival information during NMF-based deconvolution reorganizes the learned factor structure, concentrating prognostic signal into fewer, more interpretable gene programs while suppressing variance-dominant but outcome-neutral structure. This reorganization produces a more parsimonious representation of prognostically relevant biology: in PDAC, DeSurv achieved with three factors the same cross-validated concordance that standard NMF required seven to eleven factors to match, and even at each method's unrestricted optimum, DeSurv attained a higher concordance with fewer factors. The advantage is most pronounced at low factorization ranks, where standard NMF allocates its limited representational capacity across variance-dominant but prognostically inert programs — including exocrine-compositional, classical tumor, and microenvironmental variation — leaving no single factor to concentrate survival association; survival supervision corrects this allocation, enabling recovery of basal-like and microenvironmental programs even at $k = 3$. In simulations, DeSurv recovered the true factorization rank and the identity of prognostic programs more reliably than standard NMF, with the largest advantage occurring when prognostic programs explained modest variance relative to outcome-neutral signals. In PDAC, DeSurv isolated tumor-intrinsic and microenvironmental programs with clear survival associations, and these factors generalized across independent PDAC cohorts. The K= 3 selection proved robust rather than fragile: an exhaustive sensitivity analysis spanning five ranks and seven supervision strengths, with all hyperparameters fixed at their BO-selected values, confirmed that K= 3 validates adjustedly across a wide supervision range (5 of 7 α values tested; adjusted $p = 0.003$ at $\alpha = 0.55$). This robustness depended critically on the BO-selected gene-focusing parameter (ntop= 270): retaining all genes with the same regularization reduced the number of significantly validated K$\times\alpha$ combinations from 18 to 5 of 35 and eliminated K= 3 adjusted significance at the production supervision strength, establishing gene focusing as the mechanism by which survival supervision isolates a generalizable prognostic program. In PDAC, DeSurv recapitulated known tumor and microenvironmental programs, including the basal-like/classical distinction and activated versus normal stromal states. The methodological contribution lies not in the identity of these programs—which have been established through virtual microdissection (4), experimental microdissection (24), and unsupervised deconvolution (5)—but in how they are recovered. DeSurv discovers these programs de novo, without requiring pre-specified signatures or post hoc filtering, and directly quantifies their survival associations during the factorization. The resulting factors align with the consensus that has emerged from over a decade of PDAC subtyping efforts (1, 4, 7, 13), but are identified through a single optimization rather than iterative retrospective evaluation. The semi-supervised design of DeSurv reflects a deliberate balance between biological completeness and clinical relevance. The reconstruction term (weight $1 - \alpha$) penalizes deviation from the observed expression data; the survival term (weight α) directs gene programs toward outcome-relevant structure. This balance interacts with rank selection: survival supervision enables parsimonious factorizations by concentrating prognostic signal into fewer factors, so that the one-standard-error rule applied to cross-validated concordance selects smaller ranks than would be chosen by unsupervised criteria or by optimizing concordance without supervision. In PDAC, the cross-validated concordance surface was relatively flat across $k = 3$–12 for DeSurv, indicating that three supervised factors captured nearly as much prognostic information as twelve; for standard NMF, concordance increased steadily from $k = 2$ through $k = 7$–11, reflecting the need for additional factors to distribute prognostic signal that supervision would have concentrated. A K$\times\alpha$ sensitivity analysis confirmed this parsimonious selection: with production-optimal gene focusing (ntop= 270), K= 3 validated adjustedly across a broad α range, while the apparent advantage of K= 7 observed under all-gene projection disappeared entirely, demonstrating that the one-standard-error rule correctly identified K= 3 as the stable, generalizable solution. Our simulations provide further guidance on when this tradeoff is most beneficial: DeSurv's advantage is greatest when prognostic programs explain modest variance relative to outcome-neutral signals and vanishes under null conditions where no survival signal exists. DeSurv is therefore not intended to replace unsupervised NMF in all settings. When the goal is exploratory discovery without clinical endpoints, when survival data is sparse or unreliable, or when the application requires comprehensive biological characterization rather than prognostic stratification, unsupervised methods remain appropriate. The α parameter, tuned via Bayesian optimization, lets the data determine the appropriate balance for a given cancer type and cohort size, and the $\alpha = 0$ endpoint recovers standard NMF as a special case.

Several limitations should be noted. First, DeSurv assumes Cox proportional hazards, which may not hold in settings where treatment effects are delayed or non-proportional, such as immunotherapy response; extensions to more flexible survival models (e.g., accelerated failure time models or neural network-based hazard functions) are a natural direction. Second, the computational cost of Bayesian optimization over cross-validated concordance exceeds that of standard NMF and may limit scalability to very large cohorts or rapid iterative analyses; however, hyperparameter selection is performed once per dataset and the final model can then be applied to arbitrarily many validation cohorts by projection. Third, DeSurv was applied here only to PDAC, and broader benchmarking across cancer types is needed to establish generality of the variance-prognosis divergence pattern and of survival supervision as a remedy. Fourth, DeSurv's convergence guarantee (SI Appendix) applies to an idealized algorithm; the implementation includes practical modifications (backtracking, gradient clamping) that depart from the theoretical analysis, though empirically these do not affect convergence behavior. Fifth, the supervised factorization is optimized with respect to a specific clinical endpoint, and the resulting molecular programs may differ under alternative outcome definitions (e.g., progression-free vs. overall survival). Moreover, because supervision incorporates observed survival, which reflects both tumor biology and treatment received, the learned factors may implicitly encode treatment-response associations present in the training cohort. The Cox component can incorporate additional clinical covariates to adjust for known confounders during optimization, partially mitigating this concern, though residual confounding from unmeasured factors remains possible. This limits transportability to settings with substantially different treatment landscapes and means the biological in-

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terpretation of supervised factors should be understood as conditional on the therapeutic context in which they were derived. Finally, as with any supervised method, DeSurv's value depends on the quality of the outcome data: heavily censored, short follow-up, or misannotated survival information will degrade the learned programs. The nested cross-validation framework mitigates overfitting to training labels, but cannot compensate for systematic outcome misspecification.

More broadly, the principle instantiated by DeSurv—that outcome-guided dimensionality reduction targets different subspaces than variance-driven reduction—extends beyond cancer genomics. Sufficient dimension reduction theory (9) and the information bottleneck framework (25) both predict that supervised compression retains outcome-relevant structure while discarding nuisance variation. DeSurv realizes this principle within the specific constraints of NMF deconvolution, where nonnegativity preserves biological interpretability and the factorization structure enables single-sample scoring. The success of this approach in PDAC suggests that analogous frameworks may be valuable in other settings where high-dimensional measurements contain both signal and nuisance variation—including multi-omics integration, spatial transcriptomics, and electronic health records. Extending DeSurv to these domains, and to alternative outcome models beyond Cox regression, is a natural next step. As single-cell cohorts with clinical annotation grow in size, direct survival modeling at cellular resolution may complement deconvolution-based approaches, though the cohort sizes required for stable survival analysis remain available primarily in bulk expression data.

Materials and methods

A. Problem formulation and notation. Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the censoring indicators ($\delta_i = 1$ if the event is observed), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$ and linear predictor $Z^\top \beta$.

B. The DeSurv Model. DeSurv integrates Nonnegative Matrix Factorization (NMF) with penalized Cox regression to identify gene programs associated with patient survival. The method operates in a semi-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0,1]$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W,H,\beta) = (1-\alpha)\,\mathcal{L}_{\text{NMF}}(W,H) - \alpha\,\mathcal{L}_{\text{Cox}}(W,\beta), \quad [1]$$

where $\mathcal{L}_{\text{NMF}}(W,H)$ is the NMF reconstruction error and $\mathcal{L}_{\text{Cox}}(W,\beta)$ is the elastic-net penalized partial log-likelihood. A critical architectural choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\text{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients (3, 17). When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF. The Cox component also accommodates additional sample-level covariates (e.g., tumor stage or grade) alongside Z, enabling adjustment for known prognostic factors during optimization.

Optimization proceeds by alternating updates for H, W, and β, using multiplicative rules for H (3), projected gradients for W, and coordinate descent for β. Although non-convex, these updates are shown to converge to a stationary point under mild conditions (SI Appendix). Complete derivations and algorithmic details are provided in the SI Appendix.

C. Hyperparameter selection and cross-validation. Hyperparameters $(k, \alpha, \lambda_H, \lambda, \xi)$ were selected by maximizing the cross-validated C-index using Bayesian optimization, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lay within one standard error of the maximum). All model selection was performed entirely within training data using nested cross-validation; no validation cohort data were used during any stage of tuning. Because α is selected via cross-validated concordance, values large enough to overfit to cohort-specific survival patterns are penalized by poor out-of-sample performance, preventing the survival term from overwhelming the factorization. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (SI Appendix). Before validation, each column of W was truncated to its BO-selected number of top genes (details in SI Appendix), denoted $\tilde{W}$. In PDAC, Bayesian optimization selected ntop= 270 top genes per factor; a controlled K$\times\alpha$ sensitivity analysis subsequently confirmed that this gene-focusing parameter, rather than the specific λ or ν values, is the critical hyperparameter enabling adjusted external validation (see Results and SI Appendix, Tables S1–S3).

External validation was performed by projecting new datasets onto the learned programs via $Z = \tilde{W}^\top X_{\text{new}}$ and evaluating survival associations using C-index and log-rank statistics. Because the learned W is fixed at training time and validation samples are scored by simple projection, no retraining or access to validation survival data is required. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. To evaluate the quality of DeSurv-derived gene signatures for subtyping, new datasets X_{new} were clustered on genes in $\tilde{W}$, and survival differences were analyzed for the derived clusters. Further training, validation, and runtime details appear in the SI Appendix.

D. Simulation studies. Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings; each class was

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generated from a distinct gamma distribution. Sample-level factor activities H were generated from a gamma distribution.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T\tilde{W}$, where $\tilde{W}$ retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a primary scenario where prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

E. Real-world datasets. We analyzed publicly available RNA-seq and microarray cohorts of pancreatic ductal adenocarcinoma (PDAC) with corresponding overall survival outcomes. Gene expression matrices were converted to TPM, log-transformed, and filtered to remove low-expression genes. Survival times and censoring indicators were taken from the associated clinical annotations. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and the rest were used for external validation (Dijk, Moffitt, PACA, Puleo). To harmonize differences in scale across cohorts, filtered gene expression data was within-subject rank transformed before model training. More details about the datasets can be found in the SI Appendix.

F. Software and availability. An R package implementing DeSurv is available at github.com/ayoung31/DeSurv. Code and processed data used in this study are available at github.com/ayoung31/DeSurv-paper.

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Versioning

```
## DeSurv package version: 1.0.1
## DeSurv git branch: main
## DeSurv git commit: 370c88aa9a89e0c71507fbb161d226c7e2e4c61f
## Paper git branch: main
## Paper git commit: 11cf830d5c14cca1a36e410fda7073890789f038
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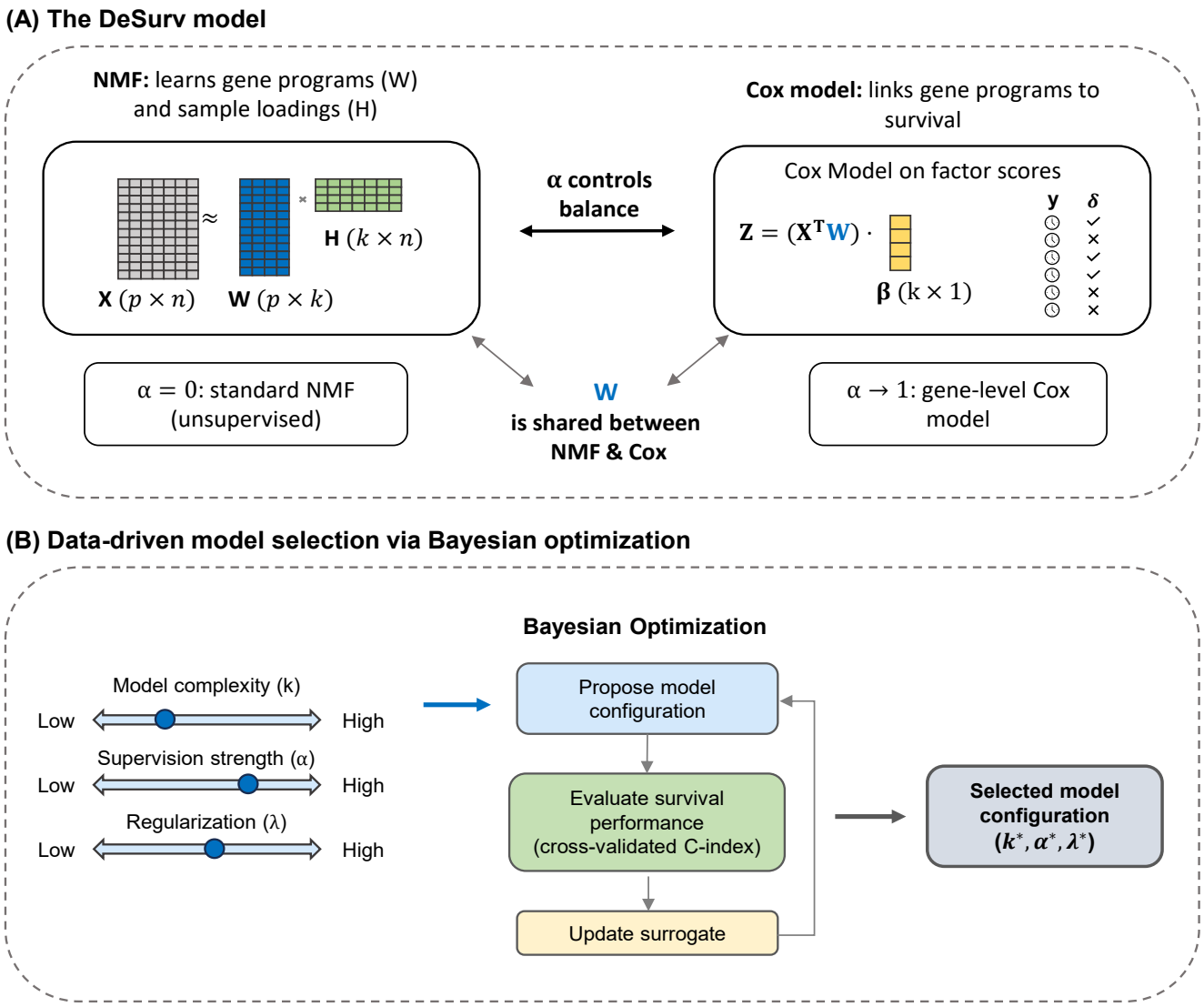

Fig. 1. Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter α controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction ($\alpha > 0$). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

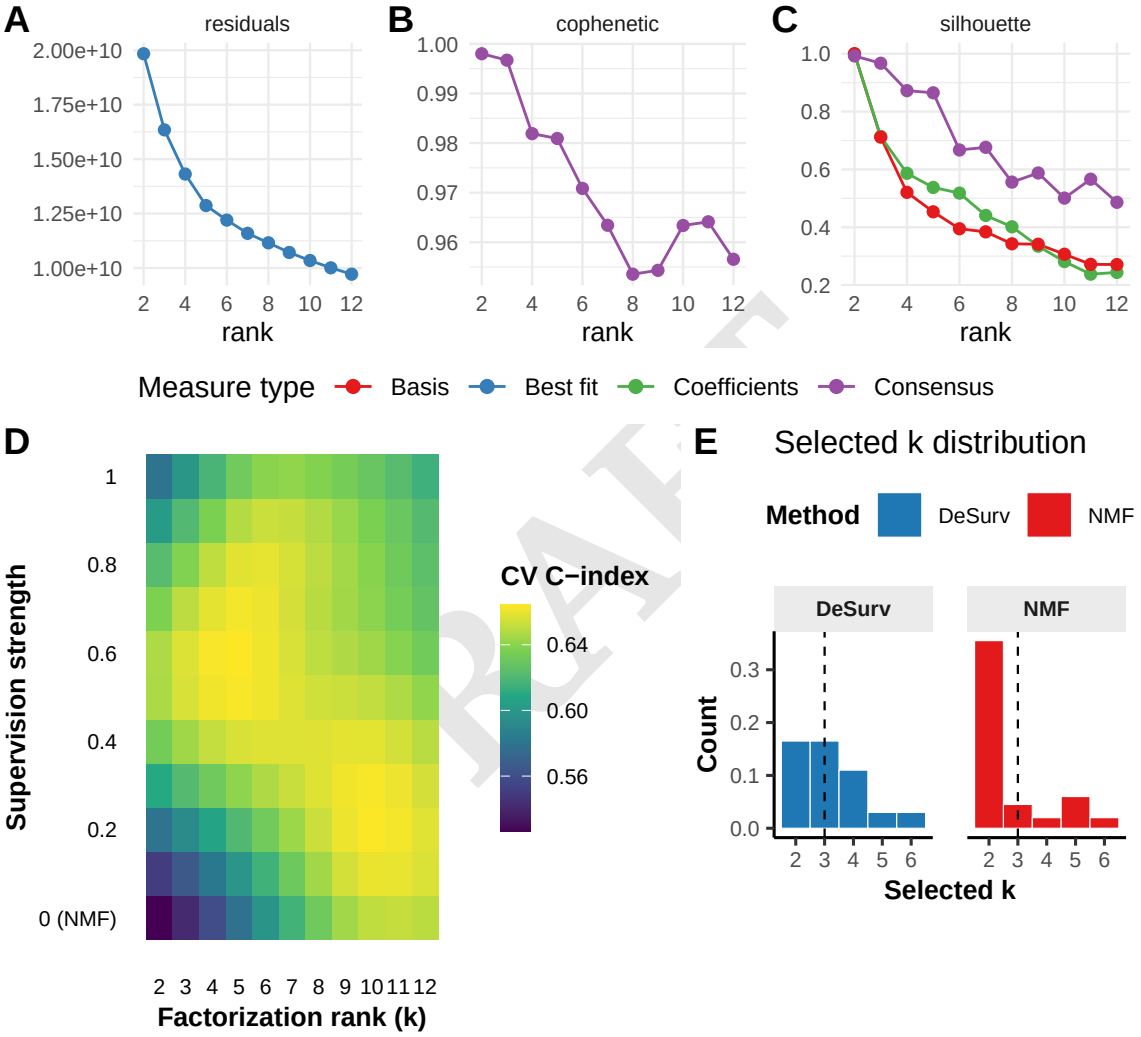

Fig. 2. (A–D) Analyses based on pancreatic ductal adenocarcinoma (PDAC) gene expression data from TCGA and CPTAC cohorts. (A–C) Standard unsupervised rank selection heuristics yield inconsistent guidance for selecting the factorization rank k. (A) Reconstruction residuals as a function of k. (B) Cophenetic correlation coefficient as a function of k. (C) Mean silhouette width across multiple distance metrics as a function of k. (D) Gaussian process predicted mean cross-validated concordance index (C-index) from Bayesian optimization over the joint space of factorization rank (k) and supervision strength (α). (E) Simulation studies with known rank ($k = 3$): distribution of selected k values across repeated replicates for DeSurv versus standard NMF with post hoc Cox modeling ($\alpha = 0$).

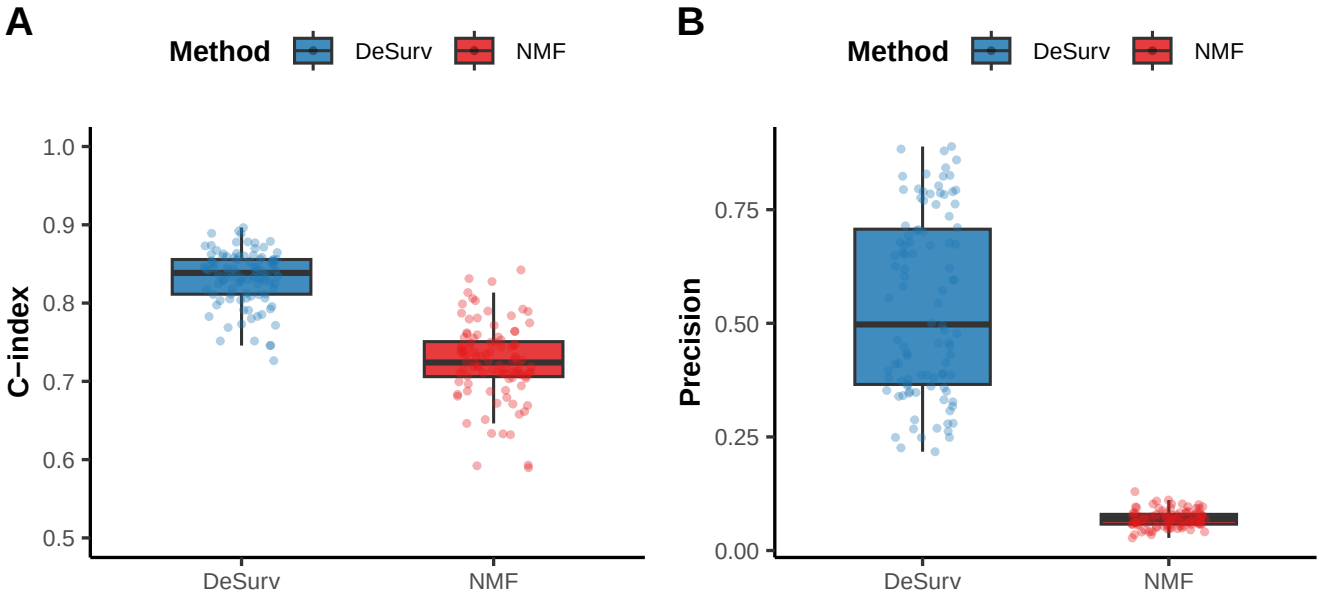

Fig. 3. Performance comparison between DeSurv and standard NMF ($\alpha = 0$) in the primary simulation scenario ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$, 100 replicates), where prognostic programs explain low variance relative to outcome-neutral programs. Both methods were tuned via Bayesian optimization over cross-validated concordance index. (A) Test-set concordance index across simulation replicates. (B) Precision (fraction of genes in a learned factor belonging to a true prognostic gene program) across simulation replicates. Each point represents one replicate.

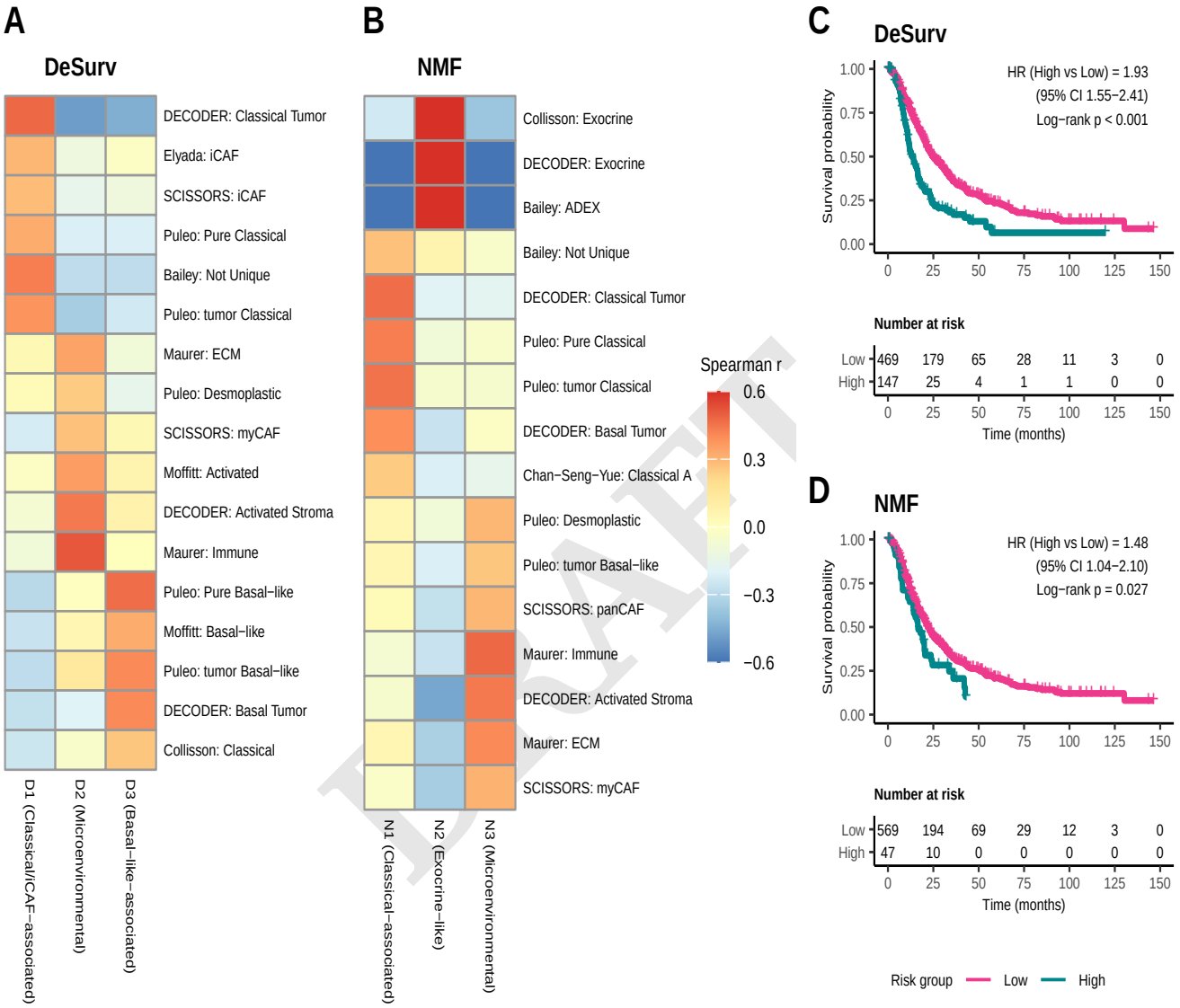

Fig. 4. Survival-informed factorization reorganizes transcriptional structure and generalizes to independent datasets. (A) Correlation of DeSurv factor gene rankings with established pancreatic ductal adenocarcinoma (PDAC) gene programs. DeSurv isolates tumor-intrinsic classical and basal-like programs and resolves an activated immune–stromal microenvironment, while exocrine-associated expression does not dominate any factor. (B) Corresponding correlations for standard NMF. NMF recovers variance-dominant structure, including an exocrine-associated factor opposing immune and stromal signatures, a classical tumor factor, and a composite microenvironmental factor. Asterisks indicate significant correlations after multiple testing correction. Only gene lists with correlation > 0.2 included in A and B. (C) Kaplan–Meier curves for pooled validation samples stratified into high and low groups using the cross-validated optimal cutpoint applied to the DeSurv linear predictor. Clear separation of survival trajectories indicates robust out-of-sample prognostic performance. (D) Corresponding analysis for the NMF-derived factor selected using the same survival-based criterion, showing weaker discrimination between risk groups.

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